

Pursuing minimally disruptive medicine: disruption from illness and health care-related demands is correlated with patient capacity

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Abstract

Background: Chronic conditions burden patients with illness and treatments. We know little about the disruption of life by the work of dialysis in relation to the resources patients can mobilize, that is, their capacity, to deal with such demands. We sought to determine the disruption of life by dialysis and its relation to patient capacity to cope.

Methods: We administered a survey to 137 patients on dialysis at an academic medical center. We captured disruption from illness and treatment, and physical, mental, personal, social, financial, and environmental aspects of patient capacity using validated scales. Covariates included number of prescriptions, hours spent on health care, existence of dependents, age, sex, and income level.

Results: On average, patients reported levels of capacity and disruption comparable to published levels. In multivariate regression models, limited physical, financial, and mental capacity were significantly associated with greater disruption. Patients in the top quartile of disruption had lower-than-expected physical, financial, and mental capacity.

Conclusions: Our sample generally had capacity comparable to other populations and may be able to meet the demands imposed by treatment. Those with reduced physical, financial, and mental capacity reported higher disruption and represent a vulnerable group that may benefit from innovations in minimally disruptive medicine. © 2016 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Keywords: Minimally disruptive medicine; Chronic care; Patient-centered care; End-stage renal disease; Dialysis

1. Introduction

Chronic conditions, defined as lasting 12 months or more and limiting self-care, independent living or require ongoing medical intervention [1], require ongoing, typically lifelong treatment that is burdensome to patients. Estimates in 2013 indicated that 117 million, or approximately half of adults in the United States, had one or more chronic conditions [2], whereas 26% of adults in the United States had multiple chronic conditions, up from 21.8% in 2001 [3]. Certain populations of patients with chronic conditions have a higher prevalence of comorbidities, such as patients with chronic kidney disease, where the prevalence of comorbidities has been estimated at 56–86%, increasing with the severity of the disease [4,5].

Chronic conditions cause pain, fatigue, and other symptoms, as well as functional limitations, which contribute to the burden of illness. Additionally, they place another burden on patients, this one arising from treatment tasks [6]. Treatment burden can be defined as treatment work, delegated by health care systems to patients and its impact on their functioning and well-being; there are growing demands on patients to organize their own care and self-manage to comply with complex regimens [7,8]. A recent systematic review found that patients with chronic conditions spent on average 86 minutes per day managing a single condition and 35 additional minutes for recommended exercise. Although not daily, patients report spending 104–151 minutes for each health care encounter, including transportation, waiting, and receiving care [9]. Further complicating chronic care is the disease-centered nature of clinical practice guidelines and quality metrics. Guidelines often fail to consider comorbidities, treatment burden, patient preferences, or patient context [10]; the consequence of their additive, uncoordinated, and often contradictory tasks is increased treatment burden [11].

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What is new?

- The Cumulative Complexity Model was published in 2012 as a conceptual model to guide research, innovation, and implementation of minimally disruptive medicine (MDM). This is the first article that seeks to provide prospective exploration of model concepts. Specifically, this article explores a population with a high workload, providing the ability to explore domains of patient capacity to enact that work. Additionally, because this patient population must adhere to treatment for survival, disruption presumably affects their ability to pursue their life's joys, which we have measured through illness and treatment intrusiveness.
- Patients on dialysis report capacity and disruption from illness and treatment that is comparable to other populations on average. However, patients most disrupted by their illness and treatment had deficits in their physical, emotional, and financial capacity.
- Chronic conditions and the burden of their treatment can overwhelm patients' lives. Clinical care should seek to identify patients that may be overwhelmed by their care and identify and address capacity issues that patients may have. Where capacity issues cannot be addressed by the health care system, treatment plans should be adapted. This requires careful clinical conversations about coping with treatment and patient capacity, of which innovations in MDM can support.

Treatment burden alone is not the only factor: Eton et al. found that multiple contextual issues may exacerbate treatment burden, such as challenges taking medications, emotional problems with family and friends, role and activity limitations, financial challenges to afford and access health care, and health care delivery inefficiencies [12]. All of these can be considered barriers to patients accessing their available abilities and resources to meet treatment needs.

Minimally disruptive medicine (MDM) accounts for these factors and is an approach to care emphasizing the “fit” of health care to patients' lives [8,13]. It is grounded in the Cumulative Complexity Model (CuCoM, see Fig. 1), a model of patient complexity that outlines how clinical and social factors complicate care and outcomes [14]. In the CuCoM, every patient has a set of demands and responsibilities in his or her life, which constitute the patient's “workload.” This workload includes a person's health care tasks, but also competing demands required for other life roles as parents, spouses, friends, caregivers, workers, and

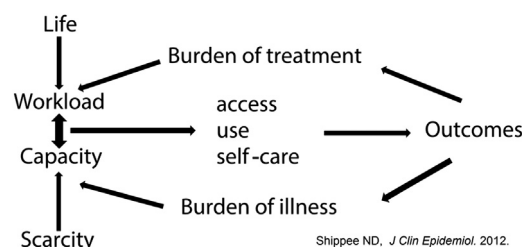


Fig. 1. Cumulative Complexity Model.

community members. In meeting these demands, patients have a finite capacity, which can be described as the abilities and resources they can mobilize to manage this workload [14]. In some ways, our theoretical understanding of capacity comports with the Conservation of Resources Theory, which states that people have objective resources, conditions, personal characteristics, and energies available; they seek to conserve these resources as they interact with their environment and loss of them may cause stress [15]. For example, to address high self-care demands, a patient may use their high health literacy to quickly and effectively make sense of complex self-care regimens and multiple follow-up visits to several physicians and solve problems; or may enroll a spouse on whom to delegate the tasks of organizing medications; or partner with a well-coordinated health care system able to support self-care.

For some patients, capacity is sufficient to meet the demands of care and the demands of life. However, others experience an imbalance—that is, when workload exceeds capacity. The CuCoM posits that this imbalance means that patients may be unable to access and use health care and perform self-care alongside the other demands in their lives [14]. Such an imbalance, we call disruption, could manifest in two ways: (1) patients prioritize life's demands, disrupting treatment adherence and self-care, and therefore resulting in worsening clinical outcome markers or (2) patients prioritize necessary clinical care and self-care at the cost of disrupted lives, manifesting as intrusion to their ability to carry out meaningful activities such as hobbies and relationships. Although some patients can afford the former with few immediate consequences, those with life-threatening conditions requiring close control (e.g., dialysis, highly active antiretroviral therapy for HIV) have little option but the latter, meaning both illness and treatment, become intrusive on their lives.

Despite some work on understanding and measuring burden of treatment [6,7,12,16–18], patient capacity remains largely unexamined. Certain domains of patient capacity have, in isolation, been shown to affect patient outcomes and behaviors [19,20]. Fatigue and self-efficacy have been shown to be associated to self-management behaviors [19–22], pain with depression status and coping ability [23], financial hardship, and social support with quality of life and mortality [24–31]. Attending to such

issues may be vital: in one systematic review, supporting patient capacity in interventions was associated with greater effectiveness in reducing 30-day hospital readmissions [32].

1.1. Dialysis, disruption, and patient capacity

These issues of workload-capacity imbalance, treatment burden, and illness burden play out in a heightened fashion for patients with end-stage renal disease (ESRD) on dialysis, making them an ideal population in which to study aspects of the CuCoM. These patients struggle with a time-consuming workload of health care and life restrictions [33,34] to decrease risks of hospitalizations and complications and increase long-term survival [35–37]. Even as dialysis patients work to meet treatment demands, they also face pain, discomfort, fatigue, nausea, insomnia, heart problems, and decreased mobility due to ESRD or dialysis itself [38–40]. These factors combine in such a way that patients on dialysis are a population that may be characterized as a population with both high illness burden and high treatment burden.

High treatment burden, combined with a low baseline capacity owing to burden of illness, leaves patients on dialysis vulnerable to disruption from workload-capacity imbalance, particularly if other areas of their capacity are reduced. As postulated in the CuCoM, this disruption may lead to problems accessing and using health care and enacting self-care. Thus, identifying patients with difficulties in varying capacity domains could identify targets for intervention and prove fruitful in ensuring patients' ability to manage the demands of their health care and a good health-related quality of life. Capacity challenges for patients on dialysis include struggles personally, emotionally, physically, socially, financially, and environmentally. Psychosocially, some patients are resilient in coping [41–43], whereas others struggle [44] and express regret, loss, and sadness related to dialysis [40]. Functional struggles and social isolation can cause both depression and anxiety [39,40], and financial stressors are common due to treatment cost and diminished wage-earning abilities [40,45,46]. However, health care services can be organized to mitigate environmental challenges (e.g., transportation and parking facilities), and those with competent staff can facilitate patient participation in decision making and learning together to empower patients and help them cope with their condition [47].

Our study aims to document a comprehensive picture of patient capacity in a group of patients on dialysis, the associations between individual domains of patient capacity, whether specific domains of patient capacity are associated with the disruptiveness of illness and treatments for patients' lives, and to provide further empirical evidence in support of the CuCoM. A key assumption in testing the CuCoM for this study is that this patient population has a high treatment workload to adapt for survival, relative to other

chronic disease populations in which treatment workload may more variable (e.g., asthma, diabetes). As such, capacity, or lack of, is assumed as the primary driver of imbalance for these patients. Additionally, because of the high degree of adherence required for self-care, disruption in this population can be measured as disruption to patients' lives.

2. Methods

2.1. Participants

We identified and invited all English-speaking patients 18 years or older presenting for hemodialysis or a home dialysis checkup to either of two dialysis units in a large multisite practice in the same Midwestern city in the United States.

2.2. Instrument development, stakeholder input, and pilot testing

We designed the survey instrument based on the CuCoM and MDM. We identified key concepts of morbidity-related and health care—related life disruption and domains of patient capacity a priori through consultation with experts in MDM and CuCoM. To identify measures, we then cross-referenced those concepts with literature searches on patient-centered experience and outcomes in dialysis; treatment demands for dialysis patients; and health-related and sociodemographic covariates (see measures in the following).

After drafting the instrument, we sought input from a patient advisory group which meets regularly to consult with researchers [48]. All patient advisors ($n = 15$) took 10–15 minutes to complete the questionnaire on their own. Primary concerns brought forth by the advisory group (e.g., the length of the instrument; sensitivity of financial capacity and income questions) were then explored during cognitive interviewing ($n = 5$) but did not require further modification to the instrument [49]. The cognitive interviewing procedure [49] included asking the patients to fill the survey aloud and included follow-up questions at the end of administration. Data from the surveys completed by patient advisors were excluded from this analysis. We also sought input from practicing clinicians (three nephrologists and two lead dialysis nurses) and a trained survey methodologist, who is expert in the field (TJB); all reviewed the survey instrument, finding it highly relevant to the patient population.

2.3. Survey administration

Following development, the lead author and eight trained research assistants approached all patients visiting the two dialysis centers during a 2-month period occurring from late January to early April 2014. The lead nurse at each site provided the research assistants a patient census

list at the beginning of the project with upcoming appointment dates. At the beginning of the day, the research assistants would identify all patients receiving dialysis that day from the list and seek to approach the next patient on the list, completing as many surveys as possible during each day as time allowed. They would confirm with the lead nurse on staff for the day that no new patients had been added, at which point they would be added to the master list. All patients were approached onsite while they received dialysis or waited for appointments. Patients were not approached if their dialysis procedure was being started or monitored actively by nursing staff, to ensure minimal burden to both patients and staff. Research assistants maintained from the master list the patients that had been approached, and whether they had completed the survey, wished to be approached again at a later time, or wished to decline participation. The research assistants administered the article-based surveys by oral interview; two patients asked to take the survey home and returned it at their next appointment. Research assistants were members of the Clinical Research Coordinator program at the clinical site; all underwent human subjects protection training and a 4-hour training on the project, survey administration, and privacy protection and oral consent procedures. The Mayo Clinic institutional review board approved the study procedures; all patients provided oral consent at the time of the survey.

2.4. Measures

2.4.1. Dependent variable

The dependent variable of interest was disruption due to illness and treatment, as measured by an adapted Illness Intrusiveness Scale, which has been validated in ESRD patients [50,51]. We chose to measure disruption as intrusiveness on life, rather than disruption of patient's health care and self-care tasks, because of the high level of adherence required for this specific population. The adapted version used here asks: "How much does your illness and/or its treatments interfere with..." and includes 13 items such as "your feeling of being healthy" and "your relationship with your spouse or domestic partner," with response options ranging from 0 to 7 and "not applicable" [52]. Not applicable answers were coded as missing. Items were summed for the scale (range: 0–91), with higher scores indicating more intrusiveness.

2.4.2. Key independent variable

Patient capacity included eight validated measures of six domains, and one scale from a [Supplemental Item Set/Appendix A](#) at www.jclinepi.com, selected as described previously. Personal capacity was measured using two scales. First, resilience was measured by the Connor–Davidson Resilience Scale 2 (CD-RISC-2) [53], previously adapted from an original 25-item version measuring "bounce-back" ability. Second, self-efficacy was measured

using the Stanford Patient Education Center's Chronic Disease Self-efficacy, Manage Disease in General Scale [52]. Physical capacity was measured with the Pain Interference and Fatigue PROMIS Short Form 4a scales [54]. Mental capacity was measured using the Emotional well-being scale from the SF-36, version 1 [55]. We assessed social capacity using the Instrumental Support PROMIS Short Form 4a scale [54]. Financial capacity was measured using the In-Charge Financial Distress/Financial Well-Being Scale [56]. Environmental capacity was measured in two ways. First, for home and neighborhood environmental capacity, we used the Physical Activity Neighborhood Environment Survey core scale, which includes items such as "It is within a 10 to 15 minute walk to a transit stop (such as bus, train, trolley, or tram) from my home" [57]. For health care environmental capacity, we used modified versions of four questions from the CAHPS Patient Health Care Home Survey [Supplemental Item Set/Appendix A](#) at www.jclinepi.com. These items were particularly aimed at understanding if the health care system was contributing to patient capacity with well-coordinated and supportive health care by asking questions such as "Thinking about your health care in the last 12 months: Did anyone in your primary care provider's office ask you if there are things that make it hard for you to take care of your health?" [58].

2.4.3. Workload

To assess workload of treatment and life demands, we included single-item questions. To assess life workload, we asked six questions such as "How many people would you say rely or depend on you for something?" For treatment workload, we asked two questions: "In a normal week, how much time do you spend [in hours] managing your health conditions?" and "How many different medications do you take daily on a regular basis?" Although ongoing work seeks to measure treatment workload and related concepts [6,7,12,59], at the time this survey was developed, such instruments did not exist. Therefore, these questions were not from previously validated instruments. For all eight questions, patients were allowed to write in the exact number and questions were treated as stand alone.

2.4.4. Covariates

General health was assessed with questions from the SF-36 version 1, such as "In general, would you say your health is": with responses ranging on a five-point scale from excellent to poor [55]. Additional covariates collected were age, gender, time on dialysis in years, marital status, employment status, household income in dollars (five categories ranging from less than 20,000 to more than 75,000), and one-way distance from the dialysis center in miles.

2.5. Analyses

All analyses were conducted in SAS, version 9.4 (NC, USA). In the descriptive statistics, we sought to

interpret the capacity relative to expected values; where comparators in chronic disease existed, the mean is compared to a chronic disease population mean from previous literature (intrusiveness, self-efficacy) [52,60–62] or in the case it did not, to the general population mean (emotional well-being, financial well-being, resilience, instrumental support, pain, and fatigue) [53,56,63,64]. Next, we analyzed Pearson bivariate correlations between all capacity scales, three workload single-item questions (those pertaining to the patient's health care workload for self or others), and intrusiveness. We then performed multivariate linear regression of intrusiveness on those capacity domains that had been found significant at the bivariate level, controlling for demographic characteristics of age, sex, and income, and the workload factors directly related to the patient's health care workload (number of prescriptions daily and hours spent managing health conditions weekly). For capacity domains found statistically significant in our multivariate linear regression model, we then compared the mean for each domain in the uppermost vs. lowermost observed quartiles of intrusiveness to assess the relationship of capacity in these domains in patients experiencing high disruption vs. low disruption. Pairwise deletion was used in the case of missing data.

3. Results

Of all potential participants approached, 137 (70%) agreed to participate (Table 1). Most patients were male, married, and retired or unable to continue to work because of their condition and treatment. Patients were on average 61 years of age and had been on dialysis for approximately 3 years.

Mean scores for each scale of capacity are reported in Table 2, whereas Fig. 2 compares the mean capacity scores of this cohort to previously collected samples. Cronbach alpha reliability was generally very good across domains. On average, patients reported capacity scores that were comparable to previous capacity domains, where comparison population data were available.

Bivariate correlations are summarized in Table 3. Self-efficacy and resilience had a weak, statistically significant, negative correlation with intrusiveness. Pain, fatigue, emotional well-being, and financial well-being had moderate, statistically significant, positive correlations with intrusiveness. Robust correlates among independent variables included self-efficacy, resilience, emotional well-being, pain, fatigue, and financial well-being (all significantly related to each other with the exception of resilience and financial well-being). Instrumental support and home and health care environment were not significantly related to intrusiveness or other independent variables.

Multivariate linear regression summarized in Table 4 illustrates that after adjusting for other factors, the capacity variables with significant associations with intrusiveness,

Table 1. Patient descriptive statistics

Characteristic	% or Mean (SD)	Missing
Female %	36.0	1
Marital status %		1
Married	51.8	
Divorced	14.4	
Widowed	11.5	
Separated	2.9	
Never been married	15.8	
Unmarried couple	3.6	
Age: mean (SD)	60.3 (15.7)	0
Time on dialysis in months: mean (SD)	32.8 (38.7)	12
Self-reported general health %		1
Excellent	5.0	
Very good	14.4	
Good	34.5	
Fair	34.5	
Poor	11.5	
Employment %		10
Employed for wages	17.6	
Self-employed	4.6	
Out of work for > 1 yr	2.3	
Out of work for < 1 yr	2.3	
A homemaker	2.3	
A student	0	
Retired	42.3	
Unable to work	28.5	
Annual household income %		18
Less than \$20,000	36.9	
\$20,000–\$34,999	23.0	
\$35,000–\$49,999	15.6	
\$50,000–\$74,999	12.3	
More than \$75,000	12.3	
One-way distance from dialysis center		3
Less than 5 miles	28.5	
6–15 miles	24.8	
16–25 miles	16.1	
26–50 miles	11.0	
More than 50 miles	19.7	

listed from strongest to weakest association (based on standardized coefficients), were as follows: fatigue, emotional well-being, financial well-being, and pain. Among covariates, only self-reported number of prescription medications and sex were significantly associated with intrusiveness. Missing income values accounts for many of the missing cases in the regression ($n = 18$). We imputed income to ensure those missing income did not differ greatly in their capacity scores, intrusiveness, or in the relationship between capacity and intrusiveness; in doing so, we found that most estimates were very similar (intercept suggested slightly lower intrusiveness—17.7 vs. 20.9, only age changed direction, and no variables changed in significance).

Table 5 summarizes the capacity domains according to the upper and lower observed quartiles of disruption to assess basic nonlinear association. In capacity domains found significant in the multivariate linear regression model, patients with the highest levels of disruption reported mean scores above the comparison mean scores for pain and fatigue, whereas their less-disrupted counterparts reported below average pain and fatigue. Similarly,

Table 2. Patient capacity descriptive statistics

Dependent variable		Mean (SD)	α	Missing
Disruption	Intrusiveness (range 0–91)	35.9 (20.9)	0.90	1
Capacity domains		Subdomains: independent variables		
Personal capacity	Resilience (possible range 0–8)	6.3 (1.7)	0.71	1
	Self-efficacy (possible range 1–10)	7.4 (2.0)	0.82	1
Physical capacity	Pain (possible range 41.6–75.6)	53.6 (11.2)	0.95	5
	Fatigue (possible range 33.7–75.8)	56.7 (11.8)	0.92	3
Emotional capacity	Emotional well-being (possible range 0–100)	74.4 (20.3)	0.84	1
Social capacity	Instrumental support (possible range 29.3–63.3)	57.4 (7.6)	0.85	3
Environmental capacity	Home environment (possible range 6–24)	13.8 (4.3)	0.61	2
	Health care environment (possible range 4–16)	13.0 (3.3)	0.69	3 ^a
Financial capacity	Financial well-being (range 1–10)	5.8 (3.0)	0.95	2

^a Twenty-two patients answered N/A to the items in this scale and were excluded from the correlation analysis.

those most disrupted reported emotional well-being and financial well-being below average to comparators, vs. their less-disrupted counterparts who reported above average in these areas. The study at hand sought to understand the ways in which with the disruptiveness of illness from and treatment are associated with specific domains of patient capacity. Table 5 illustrates that despite mean capacity scores comparable to other populations, patients most disrupted by illness and treatment face capacity limitations different from their counterparts that do not report disruption from illness and treatment.

4. Discussion

4.1. Summary of findings

As judged by this newly developed survey tool, dialysis patients in this sample reported comparable capacity in all domains and disruption to their lives by treatment and illness relative to comparison populations, in other chronic disease populations (where available) or the general population. Reduced physical, mental, and financial capacity had the most robust significant correlations with patients' reported disruptions in both univariate and multivariate analyses. Furthermore, patients with the highest disruption experienced deficits in each of these domains of capacity compared to those reporting the lowest disruption. For practice, this is important to consider: if a patient reports relatively high disruption from their illness and treatment, there may be a need to acknowledge their physical, emotional, and financial capacity, to understand how these issues play out in their ability to self-care, and to address these capacity issues in addition to traditional medical treatment alone. Because this study is the first to comprehensively assess patient capacity, these findings may provide a starting point for how and where to measure patient capacity, and similar studies should be considered in other chronic disease populations, including those with presumed lower treatment workload. The CuCoM posits that it is the balance of workload and capacity that impacts patient's abilities to access and use health care and enact

self-care. By exploring the role of capacity in patient-reported disruption in populations with varying levels of workload provides guidance on where clinicians and health professionals should target future intervention.

ESRD patients report capacity similar to other chronic disease and even general populations, despite facing severe capacity limitations and disruptive illness and treatment. There are a few plausible explanations for these findings, for example, patients may have already adapted and normalized to their circumstances. Clinicians who regularly treat outpatient dialysis patients, however, may be unsurprised by this finding. Indeed, some dialysis patients may be extremely ill and frail, yet others are enjoying life and regularly socializing despite the lengthy treatment, highlighting the heterogeneity of experience of dialysis patients and the value in measuring capacity rather than assuming that all have limitations. In addition, previously noted in the literature is the surprising resilience in this population, often underscored by cognitive styles that rely on positive reappraisal and optimism [43]. The correlation between emotional well-being and resilience in this sample may further support this observation.

Yet despite higher than anticipated capacity on average, capacity deficits were associated with disruptiveness, and patients in the quartile most disrupted by their illness and treatment showed severe limitations to their physical, mental, and financial capacity. As an indication of workload-capacity imbalance described in the CuCoM, left unchecked, high levels of disruption may portend breakdowns in self-care, poorer quality of life, and overall patient experience and the capacity deficits associated with its presence deserve attention. Further work should be done to understand the relationship between disruption, explored in this investigation, and the two resulting pathways in the model are as follows: (1) patient access and use of health care, self-care and (2) health outcomes. This will further connect elements of the CuCoM framework not addressed in the present investigation.

4.2. Implications for practice

MDM seeks to restore or work within the constraints of the balance of patient's workload and capacity, producing

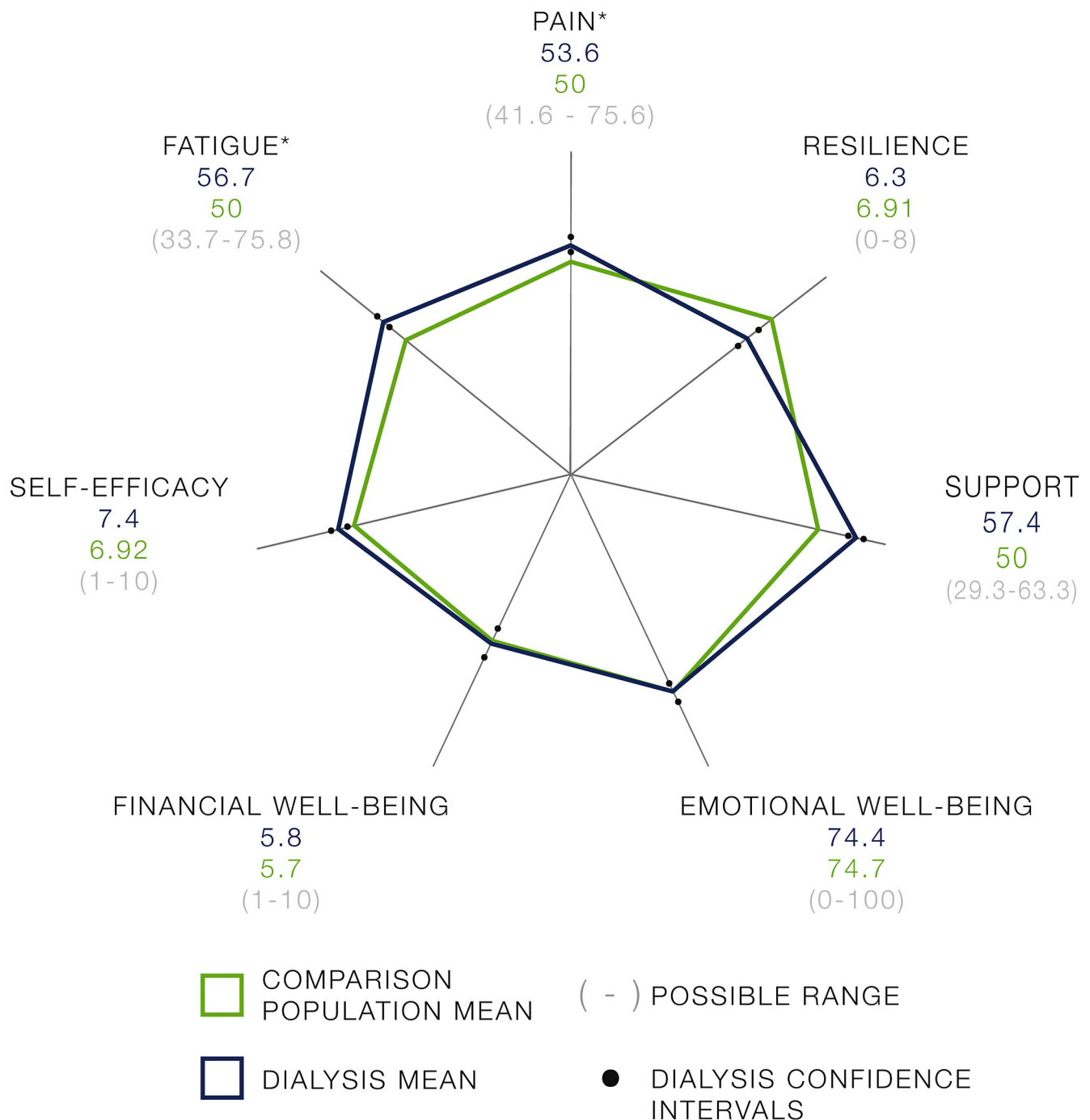


Fig. 2. Cohort capacity means vs. general capacity means. *Higher pain and fatigue scores indicate lower physical capacity.

care for chronic conditions that fits their context and produces outcomes aligned with their goals. Patient-reported illness and treatment disruption as described in this article may signal that clinical care is disrupting patients' lives, and that capacity constraints should be explored and addressed. For example, clinicians, when meeting a patient who reports high disruption from their illness and treatment should consider potential constraints related to the patient's

emotional, physical, and financial capacity. Where clinicians uncover physical or emotional problems, they may be able to address these problems medically or with referred services. Where they uncover financial constraints, they may be unable to fix the financial struggles but will need to adapt the treatment plan to ensure it does not overwhelm the patient's financial capacity. For example, clinicians and pharmacists could collaborate, in light of the

Table 3. Correlation matrix capacity variables and intrusiveness correlations

Domain	Subdomain	Intrusiveness	Self-efficacy	Resilience	Emotional well-being	Pain	Fatigue	Financial	Instrumental support	Home environment
Disruption	Intrusiveness									
Personal capacity	Self-efficacy	−0.411***								
	Resilience	−0.243**	0.287***							
Mental capacity	Emotional well-being	−0.550***	0.417***	0.385***						
Physical capacity	Pain	0.517***	−0.527***	−0.175*	−0.350***					
	Fatigue	0.612***	−0.421***	−0.257**	0.473***	0.539***				
Financial capacity	Financial well-being	−0.505***	0.404***	0.131	0.421***	−0.386***	−0.348***			
Social capacity	Instrumental support	−0.014	0.066	0.137	0.132	−0.111	−0.046	0.128		
Environmental capacity	Home environment	−0.073	0.108	−0.191*	−0.057	0.026	−0.023	0.031	0.008	
	Health care environment	0.051	0.056	−0.059	0.040	−0.049	−0.104	0.128	0.173	0.110

Missing data in Table 2.

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

patient's goals, to tailor medications regimens and prescribe treatments as necessary such that the total treatment plan does not exceed the patient's financial capacity. The CuCoM posits that could both create less burdensome care and improve patient outcomes. To our knowledge, however, MDM remains untested.

4.3. Limitations and strengths

This study should not be interpreted without considering its limitations. First, although the dialysis population was selected because of its high rates of comorbidities seen at the population level [4,5], caution must be taken when extrapolating its findings to other populations with multimorbidity. Because of the nature of dialysis treatment, the perceived illness and treatment burdens may fluctuate within a single week, which could bias measurement and extrapolation. Further study in this regard is warranted. Other sources of measurement error include the possibility that selected scales do not adequately tap into areas of patient capacity in dialysis that have the greatest deficits. For example, instrumental

support was selected as a measure of social capacity, whereas loneliness could have been explored instead. Second, potential bias in the sample should be considered; the setting in which this study occurred was a relatively socially homogenous population with good access to care at a single clinic in the Midwest. This might be responsible for certain patterns (e.g., higher than anticipated capacity across most domains) and may have limited the observed associations (due to limited variability) of environmental and social capacity domains with disruption. Findings may also differ in populations with greater heterogeneity of race, income distribution, access, providers, or geography, suggesting limited generalizability pending further research. Finally, this study was cross-sectional, and therefore, we cannot establish causality or draw longitudinal inferences. Therefore, further work is needed to test whether attention to patient capacity would causally affect disruption, and, in turn, whether this could result in better health outcomes. Moreover, confounding relationships are possible (unmeasured factors may be simultaneously causing both high disruption and low capacity). An example of such unmeasured factor could be the sudden illness of a family

Table 4. Multivariate linear regression model of patient disruption: workload and capacity variables

Domain	Variable	Estimate	P-value	Standardized β coefficient
Demographics	Intercept	20.9	0.17	
	Age	0.01	0.89	0.01
	Sex	−6.62	0.01	−0.15
	Income	1.86	0.07	0.13
Workload	Weekly hours spent caring for health	0.07	0.20	0.08
	Number of prescriptions daily	−0.66	0.03	−0.15
Personal capacity	Self-efficacy	−0.67	0.43	−0.06
	Resilience	0.08	0.91	0.01
Mental capacity	Emotional well-being	−0.25	<0.01	−0.26
Physical capacity	Pain	0.36	0.02	0.19
	Fatigue	0.65	<0.001	0.38
Financial capacity	Financial well-being	−1.64	<0.01	−0.24

Missing data, $n = 28$.

Table 5. Upper and lower quartiles of disruption across robust capacity correlates

Domain	Subdomain	Upper quartile, mean (95% CI)	Lower quartile, mean (95% CI)
Physical capacity	Pain	59.7 (55.8–63.6)	45.7 (43.0–48.3)
	Fatigue	65.6 (62.8–68.4)	48.2 (44.9–51.6)
Mental capacity	Emotional well-being	57.6 (50.3–64.9)	85.1 (80.2–90.0)
Financial capacity	Financial well-being	3.9 (3.1–4.6)	7.7 (6.9–8.5)

Abbreviation: CI, confidence interval.

member that disrupts the patient's life and also causes an acute drop in their personal capacity to enact their own care.

Despite these limitations, this study makes multiple contributions to the current literature. First, this study used previously validated scales for the measurement of the capacity domains and subjected the full survey instrument to rigorous patient and clinician stakeholder input. Additionally, the results painted the first comprehensive picture of patient capacity as described by CuCoM in any population, and the flexibility of the survey and its development process lends itself to future research in other populations with chronic conditions. Finally, this study assessed in a clinical setting a key aspect of care that is rarely examined by clinicians, and one that is beyond what can be ascertained with secondary data sources.

5. Conclusion

Patients on dialysis reported moderate disruption of their illness and treatment, and levels of capacity comparable to previous nondialysis populations. Physical, mental, and financial capacity had the most robust associations with patient-reported disruption, and patients with the highest disruption reported noteworthy capacity deficits in these three areas. Future health services research needs to be conducted to uncover the extent to which clinical teams are aware of these capacity problems, methods to routinely assess patient capacity in a minimally burdensome manner, and what associations exist between capacity deficits and patient health outcomes. Furthermore, research should seek to translate these findings by incorporating discussions of patient capacity into regular clinical conversations and testing new forms of care delivery to patients with multiple chronic conditions such as MDM.

Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jclinepi.2016.01.006>.

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